Exercise sheet 1

PRISM - Curves and Surfaces

1. Can you find a parametrization of the circle of radius r which has speed 1? What is its acceleration?
2. Consider a line given as the set of points that satisfy the equation $y = mx + b$. Find a parametrization of it which has speed 1. What is its acceleration?
3. Consider a parametrization $f : (a, b) \to \mathbb{R}^2$. Assume that its acceleration is 0. Show that it must be a straight line.
4. Suppose you are given a parametrization which has constant speed, but the speed is not 1. How can you choose a different parametrization that will have speed 1?

We will discuss the following questions during the lectures, but you way wish to think about them before that.

1. Consider a parametrization $f : (a, b) \to \mathbb{R}^2$ and two numbers c and d from the interval (a, b). How would you define the length of the curve that is between the points $f(c)$ and $f(d)$.
2. Show that if a parametrization is of constant speed, then its velocity vector and the acceleration vectors at some point are always at an angle of 90 degrees with each other.